

MAT2001 – STATISTICS FOR ENGINEERS

PRACTICE PROBLEM SHEET

1. The following table gives the average costs of a single-lens reflex camera:

800 650 300 430 560 470 640 830

400 280 800 410 360 600 310 370 Find Mean, Median, mode, standard deviation, quartile deviation, coefficient of quartile deviation, mean deviation, coefficient of mean deviation and coefficient of variation.

2. The following are the wind velocities reported at 6 P.M. on six consecutive days: 13, 8, 15, 11, 3 and 10. Find the range, mean and standard deviation.

3. Suppose two machines produce nails which are on average 10 inches long. A sample of 11 nails is selected from each machine.

Machine A: 6, 8, 8, 10, 10, 10, 10, 10, 12, 12, 14.

Machine B: 6, 6, 6, 8, 8, 10, 12, 12, 14, 14, 14.

Which machine is better than the other?

4. In a study on effectiveness of a medicine over a group of patients, the following results were obtained :

Percentage of relief	0-20	20-40	40-60	60-80	80-100
No. of patients	10	10	25	15	40

Find the mean, median, quartile deviation and standard deviation.

5. The data on ages of teachers working in a school of a city are given below:

Age (in years)	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No. of teachers	25	110	75	120	100	90	50	30

Calculate the mode, mean deviation from mean, coefficient of mean deviation, range, coefficient of range, coefficient of variation and standard deviation.

6. The data below presents the earnings of 50 labourers of a factory

Earnings (in Rs.)	1200	1300	1400	1500	1600	1800
No. of Labourers	4	7	15	12	7	5

Calculate mean deviation, quartile deviations and their coefficients.

7. Find the batting average and mean deviation for the following data of scores of 50 innings of a cricket player:

Run Scored	0-20	20-40	40-60	60-80	80-100	100-120
No. of Innings	6	10	12	18	3	1

8. Find the mean deviation from the mean, median, standard deviation and coefficient of variation of the following data :

Size of items	4	6	8	10	12	14	16
Frequency	2	5	5	3	2	1	4

9. Calculate the mean deviation for the following data of marks obtained by 40 students in a test

Marks obtained	20	30	40	50	60	70	80	90	100
No. of students	2	4	8	10	8	4	2	1	1

Wages (in Rs)	No. of persons	Wages (in Rs)	No. of persons
70-80	12	110-120	50
80-90	18	120-130	45
90-100	35	130-140	20
100-110	42	140-150	8

X:	10	20	30	40	50	60
f:	4	6	20	8	7	6

Marks	0-10	10-30	30-50	50-80	80-90	90-100
No of Students	4	12	20	8	4	2

40	25	35	30	20	40	30	20
40	10	30	20	10	5	20	

- Note : Geometric mean and Harmonic mean problems are not given
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